# 1. Introduction & System Design

The 8-puzzle is a classic sliding tile game consisting of a 3x3 grid with 8 numbered tiles and one blank space. The objective is to navigate the board from a random initial configuration to the final goal state ( `0, 1, 2, 3, 4, 5, 6, 7, 8` ). Because the state space consists of **181,440 reachable states** (half of 9!), an optimized architecture is required to solve the puzzle efficiently.

## 1.1 The "1D String" Optimization

Instead of representing the board state as a traditional 2D array ( `int[3][3]` ), we implemented it as a flat, 9-character string (e.g., `"125340678"` ).

- $O(1)$ **State Hashing:** Graph traversal algorithms require a "Visited" (Explored) set to prevent infinite looping. By using a 1D string, we can hash the board state instantly and store it in an `std::unordered_set` . Checking if a state has been visited takes $O(1)$ constant time, which is vastly superior to iterating through and comparing 2D matrices.
- **Coordinate Mapping:** Even though the board is a flat string in memory, we mathematically map it to a 3x3 grid for movement logic using:
  - **Row** = `Index / 3`
  - **Column** = `Index % 3`

## 1.2 Solvability Verification (Inversion Count)

Exactly half of all random 8-puzzle permutations are mathematically impossible to solve. If an unsolvable state is passed to the AI, it will blindly search all 181,440 states before returning a failure.
To prevent this, our engine calculates the number of **inversions** (instances where a larger numbered tile precedes a smaller one) before the search begins. If the total number of inversions is odd, the puzzle is instantly flagged as unsolvable, avoiding catastrophic CPU overhead.

## 1.3 Interactive Web Visualizer

We engineered a custom data pipeline to visualize the search output. The C++ engine exports the final solution path, runtime, and node expansions into a formatted JSON array. A custom, locally hosted HTML/JS visualizer reads this data and utilizes CSS Grid to dynamically animate the tile movements step-by-step in the browser.

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# 2. Uninformed Search Algorithms

Uninformed search algorithms (also known as blind search) explore the state space relying strictly on the graph's structure, with no estimate of how far they are from the goal state.

## 2.1 Breadth-First Search (BFS)

- **Data Structure:** `std::queue` (First-In, First-Out / FIFO).
- **Operation:** Explores the search tree level by level, ensuring all nodes at depth $d$ are checked before any nodes at depth $d + 1$.
- **Analysis:** Because it expands radially, it is **optimal** and guaranteed to find the shortest path. However, its space complexity is massive ($O(b^d)$), as it must store the entire current frontier in memory.

## 2.2 Depth-First Search (DFS)

- **Data Structure:** `std::stack` (Last-In, First-Out / LIFO).
- **Operation:** Explores a single path as deeply as possible until it reaches a dead end, then backtracks.
- **Analysis:** DFS is highly memory efficient, but it is **not optimal**. It simply returns the very first path it finds, which often results in wildly inefficient solutions (e.g., thousands of steps for a puzzle that requires 20).

## 2.3 Iterative Deepening DFS (IDS)

- **Data Structure:** A depth-limited `std::stack` wrapped in a loop.
- **Operation:** Runs a depth-limited DFS up to limit $L = 0$. If no solution is found, the limit is incremented to $L = 1$, and the search restarts from the root.
- **Analysis:** IDS perfectly merges the space efficiency of DFS with the optimality of BFS. However, its time complexity is slightly higher because it must continually discard and rebuild the upper levels of the tree.

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# 3. Informed Search (A* Algorithm)

Unlike uninformed searches, A* uses heuristics to intelligently guess the distance to the goal.

- **Data Structure:** `std::priority_queue` (Min-Heap).
- **Operation:** Nodes are popped based on the lowest total cost function: $f(n) = g(n) + h(n)$, where $g(n)$ is the cost so far, and $h(n)$ is the heuristic estimate.

## 3.1 Heuristic Admissibility & Dominance

A heuristic $h(n)$ is **admissible** if it never overestimates the true minimum cost to reach the goal ($h(n) \leq h^*(n)$). *If an admissible heuristic is used, A** is guaranteed to find the optimal path.

1. **Euclidean Distance:**
    - Calculates the straight-line distance ($h_E = \sqrt{\Delta x^2 + \Delta y^2}$).
    - By the triangle inequality, the straight-line distance is always ≤ the orthogonal grid distance. Thus, $h_E \leq h^*$.
    - **Conclusion:** Euclidean is an admissible heuristic.
2. **Manhattan Distance:**
    - Calculates the exact number of horizontal and vertical steps ($h_M = |\Delta x| + |\Delta y|$).
    - Because the 8-puzzle strictly limits movement to 4 orthogonal directions (cost = 1 per step), Manhattan perfectly estimates the minimum steps required *assuming no other tiles are blocking the path*. Thus, $h_M \leq h^*$.
    - **Conclusion:** Manhattan is an admissible heuristic.

**Dominance:**
For any two tiles on a grid, the Manhattan distance will always be greater than or equal to the Euclidean distance ( $h_M(n) \geq h_E(n)$). Because Manhattan provides a tighter, more accurate lower bound without overestimating the true cost, **Manhattan distance dominates Euclidean distance**. A dominant heuristic results in a vastly smaller search space, making A* with Manhattan the fastest and most efficient algorithm in this solver.

# 4. Experimental Benchmark Results

The following table compares the performance of all five algorithms on a randomly generated, solvable initial board state: **125340678**.

| Initial State | Algorithm | Path Cost | Nodes Expanded | Search Depth | Time (ms) |
|---|---|---|---|---|---|
| `125340678` | **BFS** | 20 | 46,513 | 20 | 26.18 |
| `125340678` | **DFS** | 7,262 | 7,421 | 7,262 | 5.41 |
| `125340678` | **IDS** | 20 | 443,150* | 20 | 254.12 |
| `125340678` | **A* (Euclidean)** | 20 | 711 | 20 | 0.94 |
| `125340678` | **A* (Manhattan)** | 20 | 427 | 20 | 0.56 |

*(Note: IDS expands a massive number of total nodes because it intentionally discards and recalculates the upper branches of the tree on every iteration to save memory).*

## 4.1 Final Conclusion

As proven by the experimental data, while BFS and IDS successfully find the optimal 20-step path, they expand tens of thousands of nodes. DFS finishes quickly but returns a wildly inefficient path of 7,262 steps.

**A* (Manhattan)** is the undisputed optimal algorithm for the 8-puzzle. Because Manhattan strictly dominates Euclidean as an admissible heuristic, it reduced the required node expansions from 46,513 (BFS) down to just 427, solving a complex 20-depth puzzle in roughly half a millisecond.